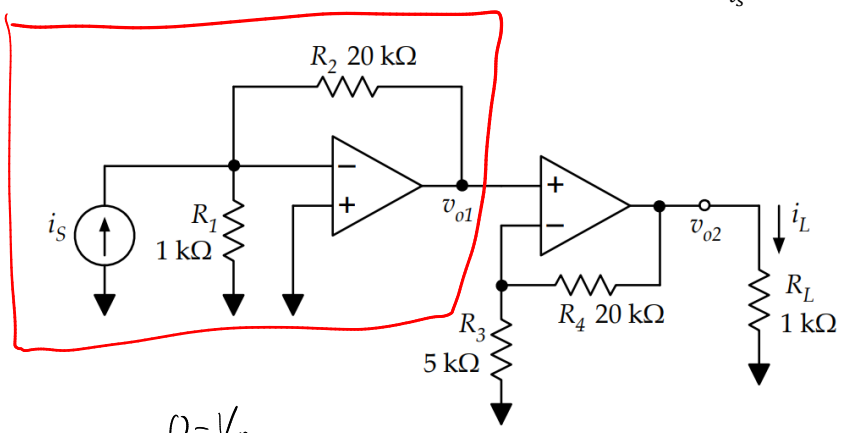
EE 2010
Fall 2026

Name Wyatt Evans

Game 26
10/5/2026

For the cascaded amplifier circuit below, calculate the current gain, $G_i = \frac{i_L}{i_s}$.



$$\frac{0 - V_{o1}}{R_2} = i_s$$

$$V_{o1} = -20\,k\Omega\, i_s$$

$$V_{o2} = \left(1 + \frac{20k}{5k}\right) V_{o1}$$

$$\frac{V_{o2}}{V_{o1}} = 1 + 4 = 5$$

$$V_{o2} = -100\,k\Omega\, i_s$$

$$i_L = \frac{V_L}{R_L} = \frac{V_{o2}}{1k} = -100\, i_s$$

$$G_i = \frac{i_L}{i_s} = \frac{-100\, i_s}{i_s} = -100$$